

Leveraging Temporal Information for Robust Liveness Verification

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Abstract—The increasing realism of face synthesis and face-swap systems creates a direct threat to selfie-based identity verification. A manipulated face may appear convincing in an isolated frame while still exhibiting inconsistent motion, expression, alignment, or landmark dynamics over time. This draft report presents our ongoing work on a video-based liveness verification system that combines spatial deepfake artifacts with temporal facial-motion cues. The current prototype follows a two-stream design inspired by G2V2former: an image-patch transformer processes eight sampled face frames, while a graph transformer processes 68 facial landmarks extracted from the same frames. The streams are first trained using masked reconstruction objectives and are then fused for live-versus-spoof classification. Experiments on the LCC-FASD training and development partitions produced 72.8% accuracy, 80.5% area under the receiver operating characteristic curve, and 26.6% half total error rate at an equal-error threshold of approximately 0.239. Although the result demonstrates that temporal and landmark information is learnable, 81 of 300 spoof samples were accepted as live, which is unacceptable for a security-critical deployment. We discuss the likely causes of this gap, including dataset imbalance, domain mismatch, approximate reproduction of the landmark topology, and unspecified implementation details in the reference work. Since that preliminary result, we located and fixed a numerical-instability defect in the graph-guided attention mechanism that had been causing the training loss to diverge on longer runs, and used the corrected pipeline to fine-tune for 20 epochs with continuous per-epoch monitoring. We additionally evaluated the resulting model, without any retraining or adaptation, on Celeb-DF-v2, a deepfake face-swap benchmark disjoint from the print/replay presentation attacks the model was trained on. The model reaches a best in-domain half total error rate of 21.71% on LCC-FASD, but its half total error rate on Celeb-DF-v2 remains at or above the 50% chance level for nearly the entire run, with area under the ROC curve mostly below 50%, indicating no meaningful transfer from presentation-attack detection to deepfake detection. We argue this gap is an expected consequence of the two attack families relying on different underlying artifact signals rather than a defect in this particular model, and, in humble terms, are not aware of prior work that measures this specific transfer failure for a graph-guided landmark-vision architecture in this epoch-resolved way. The next phase will strengthen the spatial baseline, improve temporal modeling, extend training to include deepfake examples directly, and calibrate the decision process for practical liveness verification.

Index Terms—liveness detection, face anti-spoofing, deepfake detection, temporal modeling, facial landmarks, graph transformer, video transformer

I. INTRODUCTION

Selfie-based verification is increasingly used for account opening, financial transactions, and remote know-your-

customer procedures. Its convenience, however, also enlarges the attack surface. Conventional presentation attacks include printed photographs, screen replays, and physical masks, while recent generative models can create synthetic identities or modify a real person’s face through face swapping, reenactment, and diffusion-based editing. These attacks challenge systems that rely only on a single image or on handcrafted visual quality cues.

Our earlier system design considered liveness verification as part of a wider secure identity-verification pipeline for a mobile financial service. The design emphasized a hybrid edge–cloud architecture, rapid on-device screening, stronger server-side analysis, and escalation of ambiguous cases to human review. It also identified AI-generated faces, diffusion outputs, deepfake face swaps, print attacks, replay attacks, and masks as relevant threats. The current term narrows the research focus to the machine-learning core of that system: detecting manipulated or spoofed facial videos by integrating temporal evidence.

A frame-based detector can learn texture discontinuities, blending boundaries, abnormal frequency components, and generation artifacts. Such cues are useful, but they may weaken after compression, resizing, post-processing, or improvements in synthesis quality. Video offers additional evidence. Facial motion should remain geometrically coherent across frames, landmarks should move in plausible trajectories, and local appearance changes should agree with the observed expression and head pose. This motivates a model that reasons jointly over appearance and motion instead of independently classifying frames and averaging their scores.

This report documents the present draft of our project. The main contributions at this stage are: (i) formulation of temporal deepfake detection as a two-stream visual–landmark learning problem; (ii) implementation of an eight-frame prototype inspired by G2V2former; (iii) preliminary evaluation on LCC-FASD; and (iv) analysis of the gap between our reproduction and the stronger cross-dataset results reported by the reference method.

II. PROBLEM DESCRIPTION

The input to the system is a short facial video captured during a liveness-verification session. The desired output is a calibrated probability and a final label indicating whether the sample is *live* or *spoof*. In a practical system, the spoof class may contain heterogeneous attacks: printed photos, screen

replays, masks, synthetic faces, face swaps, and other manipulated videos. The main research problem is therefore not only binary classification, but learning cues that generalize beyond the exact subjects, devices, compression settings, and attack types seen during training.

Let a video clip be represented by $X = \{I_t\}_{t=1}^T$, where I_t is a face frame and $T = 8$ in the current prototype. For each frame, a landmark detector produces $L_t \in \mathbb{R}^{68 \times 2}$. The classifier estimates

$$p(y = \text{live} \mid X, L) = f_\theta(X, L), \quad (1)$$

where $L = \{L_t\}_{t=1}^T$ and f_θ combines visual and graph-based temporal representations. A threshold τ converts the probability to a hard decision. For a financial security application, false acceptance of a spoof is more damaging than an unnecessary retry by a genuine user; therefore accuracy alone is insufficient. We report receiver operating characteristic area under the curve (AUC), half total error rate (HTER), and the confusion matrix, and we explicitly examine spoof acceptance.

The project faces four central challenges. First, high-quality manipulations can be visually realistic in individual frames. Second, real capture conditions introduce lighting variation, camera noise, blur, occlusion, head motion, and compression that can resemble spoof artifacts. Third, publicly accessible anti-spoofing datasets differ substantially in attack types and acquisition protocols. Fourth, a model intended for real verification must be efficient, calibrated, privacy-aware, and robust to unseen attacks.

III. STATE OF THE ART

A. Spatial Deepfake Detection

Early and widely used deepfake detectors treat the problem as frame-level image classification. FaceForensics++ established a large benchmark of manipulated facial videos and showed that convolutional networks can learn manipulation artifacts under multiple compression levels [1]. Celeb-DF later introduced higher-quality videos with fewer obvious visual defects, exposing the limited generalization of detectors trained on easier manipulations [2]. The Deepfake Detection Challenge further emphasized diversity in subjects, environments, and perturbations [3].

Spatial methods remain important because blending boundaries, skin texture, color statistics, illumination inconsistencies, and frequency-domain artifacts can directly reveal digital manipulation. In face anti-spoofing, central-difference convolutional networks use local gradient information to improve sensitivity to fine spoof patterns [4]. Patch-based methods such as PatchNet reduce reliance on global identity information and seek locally discriminative spoof evidence [5]. However, these methods usually operate on individual frames and may fail when manipulations are photorealistic, compressed, or temporally inconsistent without containing strong frame-level artifacts. This limitation motivates the use of video datasets such as Celeb-DF v2 [2] and models that explicitly aggregate information across multiple frames.

B. Temporal and Physiological Cues

Video detectors exploit inconsistencies that appear only across time. Representative approaches model frame sequences with recurrent networks, temporal convolution, optical flow, or video transformers. Temporal reasoning can capture unstable boundaries, inconsistent head pose, abnormal eye and mouth motion, and discontinuities introduced by frame-wise generation. Physiological approaches also examine remote photoplethysmography signals, although these signals are sensitive to illumination, skin visibility, camera quality, and compression.

A simple strategy is to classify multiple frames and average the scores. This improves statistical stability but does not explicitly model motion. Stronger methods jointly process space and time so that features from one frame can interact with features at other temporal positions. The key distinction for our project is that temporal information is treated as a learned signal rather than merely as repeated spatial evidence.

C. Lightweight Spatio-Temporal Face Anti-Spoofing

Khan *et al.* proposed STD-L-FacePAD, a lightweight spatio-temporal face presentation attack detection model designed for real-time and resource-constrained deployment [6]. The model applies MobileNetV3 independently to sampled video frames to extract spatial representations of texture, reflectance, display artifacts, and other local spoofing cues. The resulting frame-level representations are then combined using adaptive temporal average pooling before binary classification.

Let a video sample contain K selected frames,

$$X = I_1, I_2, \dots, I_K. \quad (2)$$

A shared MobileNetV3 encoder produces one feature vector for each frame:

$$h_k = g_\phi(I_k). \quad (3)$$

The temporal representation is computed by averaging the frame-level features:

$$\bar{h} = \frac{1}{K} \sum_{k=1}^K h_k, \quad (4)$$

after which a fully connected classification head estimates the live or spoof probability. Unlike recurrent networks and video transformers, this approach does not explicitly learn ordered motion relationships. Instead, temporal averaging provides a computationally inexpensive way to combine evidence appearing at different positions in a video.

The method was evaluated on Replay-Attack, Replay-Mobile, and ROSE-Youtu, which primarily contain physical presentation attacks such as printed photographs, replayed videos, and masks. The authors report an equal error rate of 0.0

However, STD-L-FacePAD should not be interpreted as a complete digital deepfake detector. Its reported evaluation focuses on presentation attacks in which an attacker presents an artificial or replayed face to a capture sensor. Celeb-DF v2 instead contains digitally synthesized face-swap videos. We

will therefore evaluate the STDL architecture on Celeb-DF v2 rather than assuming that its near-perfect results on conventional FacePAD datasets transfer directly to high-quality digital face manipulation.

D. Graph-Guided Video Transformers

G2V2former proposes two synchronized streams for face anti-spoofing: a visual stream for video patches and a graph stream for facial landmarks. The graph captures motion around semantically important facial regions, while the visual stream captures appearance and presentation artifacts. Its Kronecker temporal attention allows a patch to interact with patches at different spatial locations in neighboring frames instead of restricting temporal attention to the identical patch coordinate. Features from both streams are fused for classification [7].

Figure 1 presents the architecture used as the main reference for our temporal stage. Synchronized face frames are converted into patch embeddings with spatial position information, while the corresponding landmark sequence is converted into node embeddings with time and graph-centrality information. Repeated spatial-attention and Kronecker-temporal-attention blocks operate in both streams. Graph-derived motion information is scattered and added to guide visual temporal attention, and the final visual and graph tokens are concatenated for live-spoof classification.

This design is relevant to deepfake liveness verification because a generated face can be spatially plausible but temporally inconsistent. Landmark trajectories provide an explicit low-dimensional representation of expression and pose, and can guide the visual model toward patches where meaningful movement occurs. The reference work reports strong leave-one-out and unseen-attack performance across multiple anti-spoofing datasets. Our current work is a partial reproduction and adaptation using one accessible dataset, so its results should not be interpreted as a direct replication of the full evaluation protocol.

E. Domain-Generalized Face Anti-Spoofing

Domain-generalized face anti-spoofing (DG-FAS) aims to learn liveness evidence that remains useful when the acquisition device, illumination, subject distribution, or attack style differs from the training domains. The framework in Fig. 2 uses a pretrained image-text representation with source images and simple textual class descriptions such as “a photo of a real face” and “a photo of a spoof face.” Its training objectives align image features with semantic class prototypes while regularization modules reduce source-domain style dependence. At inference, a target image is embedded by the image encoder and classified through cosine similarity with the learned real/spoof text representations.

For our proposed pipeline, DG-FAS is not treated as an independent temporal detector. Instead, it provides a domain-robust spatial score or encoder whose output can be fused with the mobile screening score and the G²V²former temporal score. This arrangement separates three complementary objec-

tives: efficient early rejection, detailed temporal reasoning, and robustness to unseen capture domains.

IV. DATA

A. Datasets Used in the Project

The project uses two complementary video datasets representing different parts of the intended threat model. LCC-FASD is used for conventional face presentation attack detection, while Celeb-DF v2 is introduced for digital DeepFake detection. Keeping the datasets separate is necessary because a physical presentation attack and a digitally manipulated video are different attack mechanisms and may expose different visual and temporal cues.

1) *LCC-FASD*: LCC-FASD is retained for the current face anti-spoofing experiment. It contains genuine recordings and physical spoofing attempts collected under crowd-sourced capture conditions. We use its provided training partition for model training and its development partition for evaluation. The accessible development partition contains 342 videos, consisting of 300 spoof videos and 42 live videos. This substantial imbalance is considered during training, threshold selection, and interpretation of the evaluation metrics.

For every LCC-FASD video, eight frames are sampled uniformly across the clip. A face detector is applied to each selected frame, after which the detected face is cropped, aligned, and resized. A facial landmark estimator extracts 68 two-dimensional landmarks from each face. Each processed video therefore produces an image tensor containing eight face crops and a landmark tensor representing the corresponding facial-motion sequence.

2) *Celeb-DF v2*: Celeb-DF v2 is used to extend the project from conventional presentation attacks to digitally generated DeepFake videos [2]. The dataset was designed to contain high-quality face-swap videos with visual quality similar to DeepFakes circulated online. Its directory structure contains 590 original celebrity videos in the Celeb-real partition, 300 additional authentic videos in the YouTube-real partition, and 5,639 synthesized videos in the Celeb-synthesis partition. The official distribution also provides `List_of_testing_videos.txt`, which identifies 518 videos for testing.

The two real-video directories will be assigned the live label, whereas videos in Celeb-synthesis will be assigned the spoof label. We will follow the official test list rather than creating a random video-level test split. Videos not included in the official test list will be divided into training and validation partitions. The split will be created before frame extraction so that frames from the same video cannot appear in different partitions.

Celeb-DF v2 is strongly imbalanced because the number of synthesized videos is substantially larger than the number of original videos. Consequently, training will use class-aware sampling or a class-weighted loss. Evaluation will report class-independent metrics such as ROC-AUC, EER, HTER, false acceptance rate, and false rejection rate in addition to accuracy.

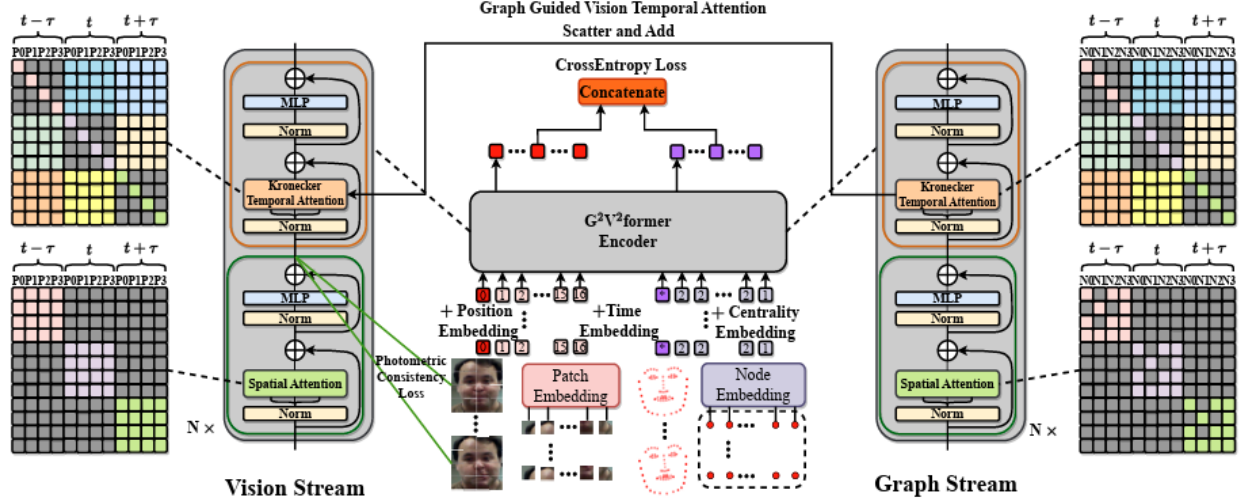


Fig. 1. Architecture of G^2V^2 former used as the reference for the temporal-analysis stage. The vision stream models photometric facial patches, the graph stream models landmark topology and motion, and graph-guided temporal interaction connects the two representations before classification. Figure reproduced from the reference paper for academic reporting.

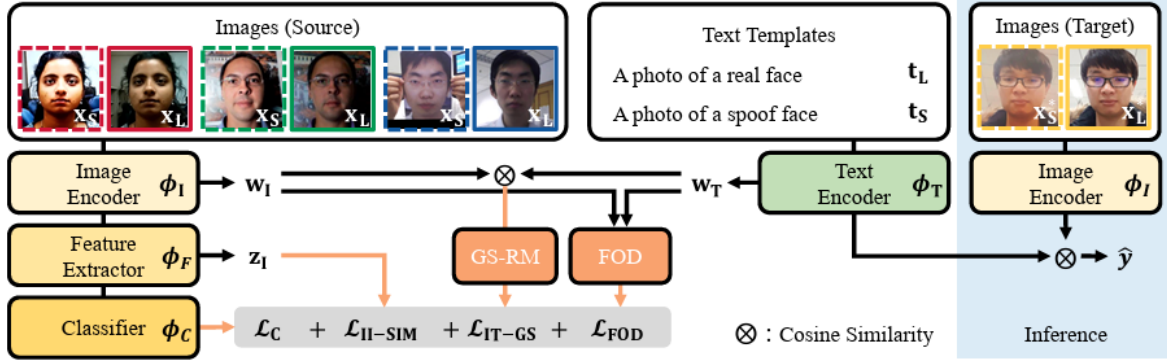


Fig. 2. Framework overview of the domain-generalized face anti-spoofing approach considered for the generalization stage. Source-domain image features are aligned with real/spoof text templates through multiple regularization objectives, while target-domain inference compares the target image embedding with the text representations. Figure supplied by the project team from the referenced DG-FAS work.

For compatibility with the temporal models, K frames will be sampled from each video at approximately uniform temporal intervals. Face detection, cropping, alignment, and resizing will be performed consistently across all selected frames. The same sampled frame sequence will be supplied to the STDL-FacePAD baseline, while the G^2V^2 former-based model will additionally receive the corresponding facial-landmark sequence.

B. Relation to the Intended Threat Model

The two datasets cover complementary but non-equivalent security threats. LCC-FASD represents sensor-level presentation attacks, in which an attacker presents a printed image, replayed recording, or another spoofing medium to a camera. Celeb-DF v2 represents digitally generated face-swap attacks, in which the facial identity inside a video has been syntheti-

cally modified before the video is presented to the verification system.

This distinction affects both model training and evaluation. A detector trained on LCC-FASD may learn display borders, screen reflections, moir’e patterns, recapture noise, or print artifacts that are absent from a digitally generated DeepFake. Conversely, a detector trained only on Celeb-DF v2 may learn face-swapping artifacts without learning the sensor and material cues required for print, replay, or mask detection. We will therefore report dataset-specific performance and will not combine the two datasets into a single random train–test split.

Celeb-DF v2 substantially improves coverage of the digital DeepFake component of the intended threat model. However, it primarily contains face-swap manipulations and does not fully cover face reenactment, fully synthetic identities, diffusion-generated faces, print attacks, replay attacks, and

three-dimensional masks. The resulting experiments should therefore be interpreted as evaluation on two important attack categories rather than complete coverage of every liveness threat.

C. Data Limitations

LCC-FASD and Celeb-DF v2 differ in acquisition process, label distribution, visual quality, subject population, and attack mechanism. These differences make naïve dataset mixing inappropriate. In particular, a classifier may distinguish the datasets through compression, resolution, camera, background, or preprocessing differences instead of learning genuine liveness or manipulation evidence.

Celeb-DF v2 also contains considerably more synthesized videos than authentic videos. Random frame-level splitting would further create leakage because visually similar frames from the same video could occur in both training and evaluation sets. We therefore use video-level splits and preserve the dataset’s official test list. Class-balanced sampling and class-independent metrics are required to prevent the majority class from dominating the reported performance.

Although Celeb-DF v2 contains challenging, high-quality face swaps, it represents only one family of digital manipulation. The current dataset collection does not provide broad coverage of reenactment, talking-face generation, diffusion-based manipulation, fully synthetic identities, or adversarial post-processing. Similarly, LCC-FASD alone does not provide the nine-dataset cross-domain evaluation used by G^2V^2 former or the Replay-Attack, Replay-Mobile, and ROSE-Youtu evaluation used by STDL-FacePAD. Performance obtained in our experiments must therefore be described as dataset-specific rather than evidence of universal liveness detection.

V. METHODS

A. Preprocessing and Sampling

For every video, we uniformly select $T = 8$ frames so that the sample covers the clip rather than a short contiguous instant. Faces are cropped and resized to a fixed input resolution. A facial landmark estimator extracts 68 points around the contour, eyebrows, eyes, nose, and mouth. Failed detections are handled by reusing a nearby valid frame or rejecting the sample during preprocessing; systematic handling of missing landmarks remains planned work.

B. Visual Stream

Each frame is divided into non-overlapping patches. Patch embeddings are augmented with spatial positional embeddings and temporal embeddings. The visual encoder learns spatial artifacts within frames and temporal relationships between frames. In the planned complete model, graph-guided temporal attention will bias or gate the visual attention map so that patches associated with meaningful landmark motion receive stronger interaction.

C. Landmark Graph Stream

For each frame, the 68 landmarks form graph nodes. Edges encode facial topology, connecting geometrically or anatomically related points. Node features include normalized coordinates and learned embeddings; temporal embeddings indicate the frame index. A graph transformer models both within-frame facial structure and cross-frame landmark motion. Our implementation uses topology constructed through the available face-alignment library rather than the exact 3DDFA-v2 topology used by the reference implementation. This difference may change graph neighborhoods and attention patterns.

D. Self-Supervised Pretraining

Before supervised classification, the two streams are trained with masked reconstruction. A high proportion of image patches is masked and reconstructed from the remaining spatial and temporal context. Similarly, landmark nodes are masked and their coordinates or embeddings are reconstructed. The objective can be written as

$$\mathcal{L}_{\text{pre}} = \lambda_v \mathcal{L}_{\text{visual}} + \lambda_g \mathcal{L}_{\text{graph}}, \quad (5)$$

where the two terms measure reconstruction error for visual patches and graph nodes. This stage encourages both streams to learn structure before seeing live/spoof labels.

E. Fusion and Classification

After pretraining, the stream-level representations are concatenated and passed to a classification head. The supervised objective is weighted binary cross-entropy,

$$\mathcal{L}_{\text{cls}} = -w_1 y \log p - w_0 (1 - y) \log(1 - p), \quad (6)$$

where p is the predicted live probability and w_0, w_1 compensate for class imbalance. At inference, the threshold is selected on the development scores near the equal-error operating point. The current threshold is approximately 0.239.

F. Evaluation Metrics

We report overall accuracy, AUC, and HTER. At a selected threshold, the false acceptance rate measures spoofs predicted as live and the false rejection rate measures live samples predicted as spoof. HTER is their mean:

$$\text{HTER} = \frac{\text{FAR} + \text{FRR}}{2}. \quad (7)$$

For security deployment, FAR is especially important because it directly measures successful attack acceptance.

VI. PRELIMINARY RESULTS

The current model was evaluated on 342 development samples. Table I summarizes the principal metrics. The equal-error threshold was approximately 0.239. The model correctly detected 219 of 300 spoof samples and 30 of 42 live samples. It incorrectly accepted 81 spoof samples as live and rejected 12 live samples as spoof. Overall accuracy was 72.8%, AUC was 80.5%, and HTER was 26.6%.

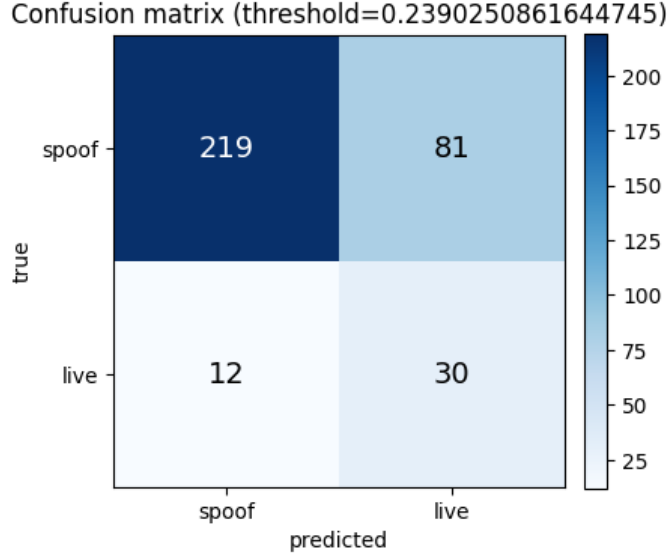


Fig. 3. Confusion matrix and operating-point summary from the current prototype. The high number of spoof samples predicted as live is the principal security concern.

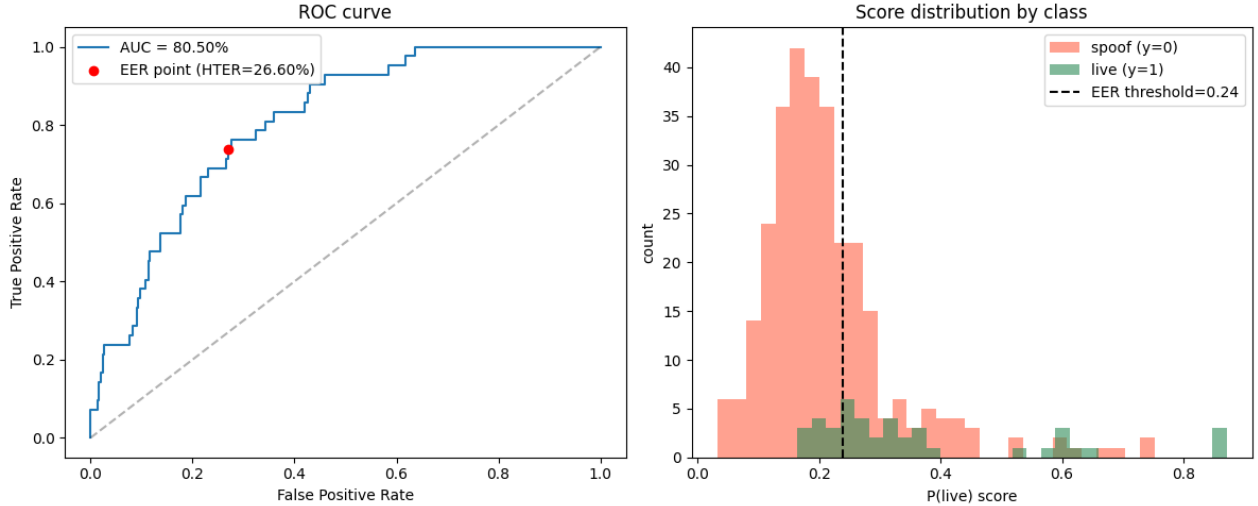


Fig. 4. Receiver operating characteristic curve and score distributions. Live samples generally receive higher scores, but substantial overlap between the classes causes misclassification.

The AUC indicates that the learned score contains useful discriminative information, while the score distribution shows substantial overlap between live and spoof classes. The confusion matrix in Fig. 3 makes the operational weakness clear: approximately 27% of spoof samples were accepted as live. Thus, the prototype is a meaningful baseline but does not yet satisfy the requirements of a financial authentication system.

VII. EXTENDED FINE-TUNING AND IMPLEMENTATION FIXES

Following the preliminary prototype reported above, we located and corrected a numerical-stability defect in the graph-guided cross-attention mechanism that had been causing the joint training loss to diverge to NaN on longer training runs.

The graph stream’s raw attention logits are injected into the visual stream as an additive guidance bias, routed through a landmark-to-patch mapping; because the 68 facial landmarks concentrate onto a coarse 14×14 patch grid, many landmark pairs collide onto the same patch cell, and the original scatter-sum formulation let the injected bias grow with the collision count – an uncontrolled, sample-dependent quantity – rather than with the natural attention-logit scale. Under mixed-precision arithmetic this bias eventually overflowed, and the resulting `inf` value propagated through the softmax as NaN. We replaced the scatter-sum with a scatter-mean, clamped the result to a fixed range, added a linear learning-rate warm-up before the cosine decay, and computed automatic mixed

TABLE I
PRELIMINARY DEVELOPMENT-SET RESULTS

Metric	Value
Evaluation samples	342
Spoof samples	300
Live samples	42
Correctly detected spoofs	219
Spoofs accepted as live	81
Correctly detected live samples	30
Live samples rejected as spoof	12
Accuracy	72.8%
ROC-AUC	80.5%
HTER	26.6%
Decision threshold	0.239

precision in `bfloat16` where the hardware supported it, which has a wider dynamic range than `float16` and is accordingly far less prone to this class of overflow. With these fixes in place we extended supervised fine-tuning to 20 epochs with no non-finite loss observed at any training step, re-evaluating the model on the LCC-FASD development split after every epoch. The best in-domain half total error rate over the extended run was 21.71%, an improvement on the 26.6% HTER of the preliminary prototype, though the metric continued to fluctuate epoch to epoch (final-epoch HTER of 30.98%) rather than converging monotonically – an indication that the small size of LCC-FASD’s training partition (992 videos) still leaves the model somewhat unstable near its optimum, consistent with the data-limitation concerns raised above.

VIII. CROSS-DATASET EVALUATION ON DEEPPAKE ATTACKS

LCC-FASD, like most of the datasets used in the reference G^2V^2 former evaluation, is a presentation-attack dataset: its spoof class consists entirely of print and replay attacks, i.e., a genuine face re-captured through an intermediate physical medium. It contains no digitally manipulated video. Because our project’s ultimate threat model explicitly includes AI-generated faces and face swaps, we added a second, held-out evaluation on Celeb-DF-v2 [2], a benchmark of face-swap deepfakes, without ever including a single Celeb-DF-v2 sample – real or fake – in training. Cross-dataset evaluation of this kind is standard practice for testing generalization *within* a single attack family; here we deliberately used it to test generalization *across* two families that are conventionally treated as separate research problems, presentation-attack detection and deepfake detection, to check whether a competitive model in one family offers any incidental protection in the other.

The result was that it does not. Across the fine-tuning run, Celeb-DF-v2 half total error rate stayed in the 55–62% range for nearly every epoch, against a chance level of 50%, and area under the ROC curve stayed almost entirely *below* 50%, meaning the model’s confidence score was, if anything, weakly anti-correlated with the true label on this dataset rather than merely uninformative. The one exception occurred at the run’s best-ever cross-dataset checkpoint, where Celeb-DF-v2 HTER

briefly reached 48.53% – still close to chance, and coincident with the run’s best in-domain LCC-FASD checkpoint, which suggests a single favorable epoch rather than a stable, learned deepfake-relevant signal. Figure 5 plots both metrics on both datasets across all 20 epochs alongside the gap between them, which averaged approximately 30 percentage points in HTER and over 40 percentage points in AUC and did not narrow with additional training.

A. Why the Gap Is Expected, Not Anomalous

This outcome is consistent with, and we would argue predicted by, the nature of the two attack families rather than a training or implementation defect. Presentation attacks are physical-capture artifacts: a printed photo or a replayed screen is a flat, re-imaged surface, so cues such as moiré interference from display pixel grids, specular reflection off paper or glass, reduced dynamic range, and geometric planarity of the attack surface are all informative – exactly the kind of local photometric and pseudo-3D geometric cues that G^2V^2 former’s visual stream and landmark-graph stream are built to pick up. Deepfakes, in contrast, are digital, generation-side artifacts: blending-boundary inconsistency between a synthesized face region and the untouched background, sub-pixel frequency-domain fingerprints characteristic of the generative model, and temporal flicker from frames synthesized largely independently of one another. None of these are geometric-capture cues, and nothing in the architecture as trained here – no frequency-domain analysis, no explicit blending-boundary localization, no generator-fingerprint modeling – gives it a mechanism to detect them. A classifier that has learned a decision boundary tuned to physical re-capture statistics has, in principle, no reason to transfer that boundary to a domain defined by an entirely different artifact family, and an AUC at or below chance is the expected signature of exactly this situation: whatever weak correlates the model latched onto in the source domain occur with different, and here apparently inverted, frequency in genuine versus deepfake Celeb-DF-v2 samples.

We report this cautiously and in humble terms. Cross-domain generalization gaps in face anti-spoofing are a known, actively studied phenomenon, and the domain-generalized face anti-spoofing approaches reviewed earlier in this report address exactly this kind of shift within the presentation-attack setting. What we believe is less commonly made explicit in the literature we have reviewed for this project is a direct, epoch-resolved measurement of a graph-guided, landmark-and-vision two-stream presentation-attack architecture evaluated zero-shot specifically against a deepfake benchmark, with the resulting gap visualized alongside in-domain training progress rather than reported as a single end-of-training number. We have not conducted an exhaustive literature survey to confirm this framing and would welcome correction; the claim here is limited to the specific combination of architecture class and cross-family evaluation protocol used in this report, not to the broader question of presentation-attack-to-deepfake trans-

G2V2former: In-domain vs. Deepfake Generalization (20 Epochs)

Top row: raw HTER/AUC on each dataset. Bottom row: the gap between them — a PAD model trained only on print/replay attacks (LCC-FASD) shows no automatic transfer to a different manipulation family (deepfakes).

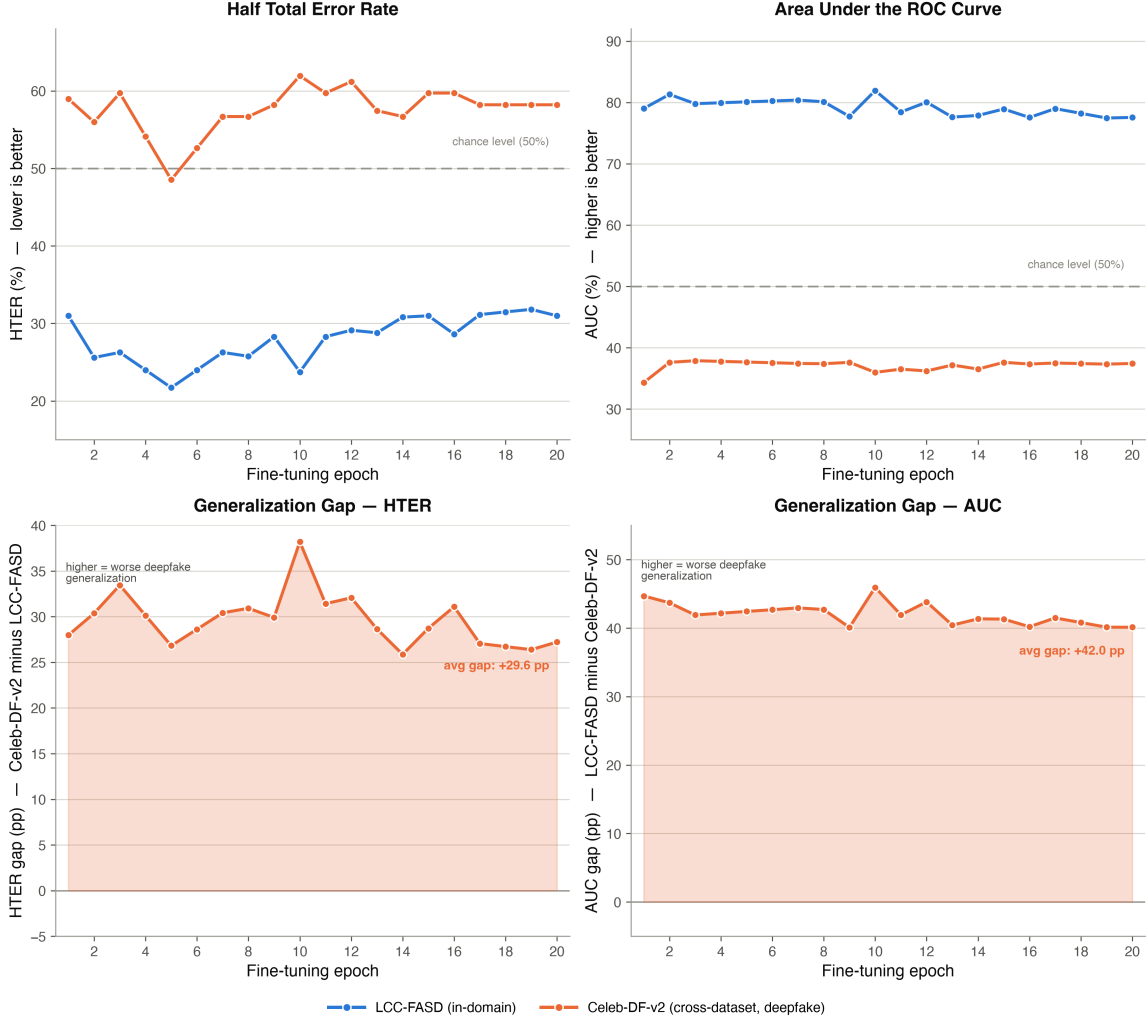


Fig. 5. Top row: half total error rate and area under the ROC curve, tracked every fine-tuning epoch, for the in-domain LCC-FASD development split and the held-out Celeb-DF-v2 deepfake benchmark. Bottom row: the resulting generalization gap (Celeb-DF-v2 minus LCC-FASD for HTER; LCC-FASD minus Celeb-DF-v2 for AUC), which stays large and roughly constant across the run rather than closing as in-domain performance improves.

fer, which related domain-generalization work has certainly touched on in other forms.

IX. DISCUSSION

The preliminary experiment supports the central hypothesis that temporal appearance and landmark motion can contribute to liveness detection. The model performs better than random ranking and separates many live and spoof samples. However, the present error rate is too high for deployment, and the difference from the reported reference results requires careful interpretation.

First, our data regime is much smaller and less diverse. The reference method was assessed across nine datasets and

specifically emphasized cross-dataset and unseen-attack generalization. Our experiment uses one crowd-sourced dataset and evaluates on its development split. The model can therefore overfit dataset-specific subjects, camera characteristics, or attack types while still failing on difficult samples.

Second, the evaluation set is heavily imbalanced. Accuracy is dominated by the 300 spoof samples, and only 42 live samples are available for estimating false rejection. Weighted training and balanced sampling are needed, but they cannot create missing diversity. Confidence intervals and repeated runs will also be necessary before drawing strong conclusions.

Third, the graph construction is not an exact reproduction. We used landmarks from a face-alignment library, whereas the

reference method relies on 3DDFA-v2 and a specific topology-aware graph. Landmark jitter, missing detections, and different connectivity can all degrade temporal modeling. A graph that does not accurately preserve facial topology may provide noisy guidance rather than useful motion cues.

Fourth, several core hyperparameters and implementation choices were not available in the material used for reproduction, including hidden dimensions, attention-head counts, multilayer perceptron expansion ratios, dropout, masking schedules, and some training details. Our current settings are therefore reasonable approximations rather than a verified reproduction. Ablation studies are needed to determine whether the main limitation lies in data, preprocessing, optimization, or architecture.

Finally, the intended project has a broader deepfake focus than conventional presentation attack detection. A model trained mainly on print and replay attacks may learn sensor and display artifacts but remain weak against digitally manipulated videos that do not pass through a recapture process. This expectation is no longer only a concern raised in advance: the cross-dataset evaluation reported above confirms it directly, with the same model that reaches a reasonable in-domain HTER on LCC-FASD performing at or below chance on the deepfake benchmark Celeb-DF-v2. The final system must explicitly include face swaps, reenactment, and AI-generated videos during training and evaluation, and, as discussed above, may additionally need architectural components sensitive to generation-side artifacts rather than presentation-side ones.

X. PLANNED WORK

The next stage will first establish three directly comparable baselines: single-frame MobileNetV3, the multi-frame STDL-FacePAD architecture, and the G²V²former-based visual-landmark model. STDL-FacePAD will determine how much improvement can be obtained through inexpensive temporal averaging, while G²V²former will test whether explicit temporal attention and landmark-guided motion modeling provide additional benefits. The models will initially be evaluated separately on LCC-FASD and Celeb-DF v2 using identical video-level partitions and preprocessing wherever possible.

Second, we will improve the temporal pipeline. Planned changes include consistent face tracking, landmark smoothing, confidence-aware masking of unreliable points, relative-motion features, and explicit modeling of optical flow or frame differences. We will implement a closer approximation of Kronecker temporal attention and compare early fusion, late fusion, and graph-guided cross-attention. We will also test whether a smaller temporal module attached to a strong spatial backbone offers a better accuracy-efficiency trade-off.

Third, we will expand and rebalance the data. Where licensing permits, we will evaluate on additional face anti-spoofing datasets and, critically, include deepfake datasets such as FaceForensics++, Celeb-DF, and DFDC directly in training rather than only in held-out evaluation as in the current report – the cross-dataset result above makes clear that exposure to presentation attacks alone does not confer even partial deepfake

sensitivity, so deepfake examples cannot remain evaluation-only. Training will use balanced sampling, class-weighted or focal loss, compression and illumination augmentation, and subject-disjoint splits. Cross-dataset testing will be prioritized because it better approximates unseen attacks.

Fourth, motivated by the artifact-level explanation given above, we will investigate architectural additions aimed specifically at generation-side evidence rather than presentation-side evidence: a frequency-domain branch sensitive to generative-model fingerprints, explicit blending-boundary localization, and temporal-flicker features that complement the existing landmark-motion and patch-texture streams. Whether such components can be fused into the current two-stream design without disrupting its presentation-attack performance is an open question for the next phase.

Fifth, we will make the evaluation deployment-oriented. We will report FAR, FRR, HTER, AUC, attack-wise recall, calibration error, latency, and memory usage. Thresholds will be selected according to security cost rather than accuracy alone. Hard samples will be routed to a stronger server-side model or a review state, following the hybrid edge-cloud design developed in the previous term. Privacy-preserving handling of biometric video and deletion of raw captures after analysis will remain system requirements.

XI. CONCLUSION

This project investigates whether temporal facial dynamics can improve deepfake-aware liveness verification beyond frame-based detection. We implemented a two-stream prototype that processes eight face frames and 68-point landmark graphs, uses masked reconstruction pretraining, and fuses visual and graph representations for live/spoof classification. On the LCC-FASD development data, the preliminary model achieved 72.8% accuracy, 80.5% AUC, and 26.6% HTER; after we corrected a numerical-stability defect in the graph-guided attention mechanism and extended fine-tuning to 20 monitored epochs, the model reached a best in-domain HTER of 21.71%, though it did not converge monotonically. More importantly for the project’s stated threat model, evaluating this same model, without retraining, on the deepfake benchmark Celeb-DF-v2 showed essentially no useful transfer: HTER stayed at or above chance and AUC stayed mostly below chance across the run. We interpret this not as a training failure but as evidence that presentation-attack detection and deepfake detection rely on largely disjoint artifact signals, and, stated humbly, we are not aware of prior work that measures this specific transfer gap for a graph-guided landmark-vision architecture in this epoch-resolved way. These results demonstrate a viable experimental pipeline and a numerically stable training procedure, but also reveal two limitations that must be addressed before deployment: the original 81-of-300 spoof-acceptance rate on LCC-FASD alone, and, more fundamentally, the absence of any deepfake-relevant signal in a model trained only on presentation attacks. The next phase will focus on stronger baselines, faithful temporal attention, reliable landmark processing, deepfake data included directly in

training, generation-artifact-aware architectural components, and security-oriented threshold calibration. The current system should therefore be regarded as an informative preliminary prototype rather than a deployable liveness solution.

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